

Visibility Metrics for Java

References

- [Visibility Metrics and the Importance of Hiding Things](#)
- [Calculate metrics](#)
- [Controlling Access to Members of a Class](#)
- [Neo4j Python Driver](#)

Relative Visibility Of Types

A Java class or interface may be declared with the modifier `public`, in which case it is visible to all classes everywhere. If a class or interface has no modifier (the default, also known as package-private), it is visible only within its own package.

The relative visibility is the number of inner components that are visible outside (public) divided by the number of all types:

$$relativevisibility = \frac{public\ types}{all\ types}$$

Using package protected types is one of many ways to improve encapsulation and implementation detail hiding.

How to apply the results

The relative visibility is between zero (all types are package protected) and one (all types are public). A value lower than one means that there are types that are declared package protected. The lower the value is, the better implementation details are hidden.

Non public classes can't be accessed from another package so they can be changed without affecting code in other packages. They clearly indicate functionality that only belongs to one package. This also motivates to use more classes and to split up code into smaller pieces with a single responsibility and reason to change.

Table 1a - Top 40 artifacts with lowest median of package protection encapsulation

This table shows the relative visibility statistics aggregated for all packages per artifact and focusses on artifacts with many packages and hardly any package protected types (lowest median, high visibility). Package protected types would help to improve encapsulation.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Global_relative_visibility_statistics_for_types

	artifact	all	public	min	max	average	percentile25	percentile50	percentile75	percentile90
0	axon-common-5.0.3	156	129	0.615385	1.000000	0.893102	0.791304	1.000000	1.000000	1.000000
1	axon-spring-boot-autoconfigure-5.0.3	75	53	0.512195	1.000000	0.887081	0.848684	1.000000	1.000000	1.000000
2	axon-conversion-5.0.3	35	32	0.700000	1.000000	0.940000	1.000000	1.000000	1.000000	1.000000
3	axon-update-5.0.3	23	23	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
4	axon-tracing-opentelemetry-5.0.3	5	5	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
5	axon-modelling-5.0.3	92	84	0.692308	1.000000	0.901923	0.860577	0.900000	1.000000	1.000000
6	axon-messaging-5.0.3	579	439	0.250000	1.000000	0.802085	0.727273	0.842105	1.000000	1.000000
7	axon-server-connector-5.0.3	72	58	0.642857	0.892857	0.788495	0.727273	0.833333	0.846154	0.874176
8	axon-test-5.0.3	78	63	0.600000	1.000000	0.809768	0.726562	0.791667	0.925725	0.978261
9	axon-metrics-micrometer-5.0.3	16	12	0.727273	1.000000	0.825758	0.738636	0.750000	0.875000	0.950000
10	axon-eventsourcing-5.0.3	100	66	0.200000	0.900000	0.599721	0.480769	0.666667	0.734921	0.813333

Table 1b - Top 40 artifacts with highest median of package protection encapsulation

This table shows the relative visibility statistics aggregated for all packages per artifact and focusses on artifacts with many packages and the highest median of package protected types (low visibility). Package protected types help to improve encapsulation.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Global relative visibility statistics for types

[illegible]

Table 1 Chart 1 - Relative visibility in artifacts

```
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_  
matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Para  
meters 'cmap' will be ignored  
    scatter = ax.scatter(  
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_  
matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Para  
meters 'cmap' will be ignored  
    scatter = ax.scatter(  
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_  
matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Para  
meters 'cmap' will be ignored  
    scatter = ax.scatter(  
<Figure size 640x480 with 0 Axes>
```

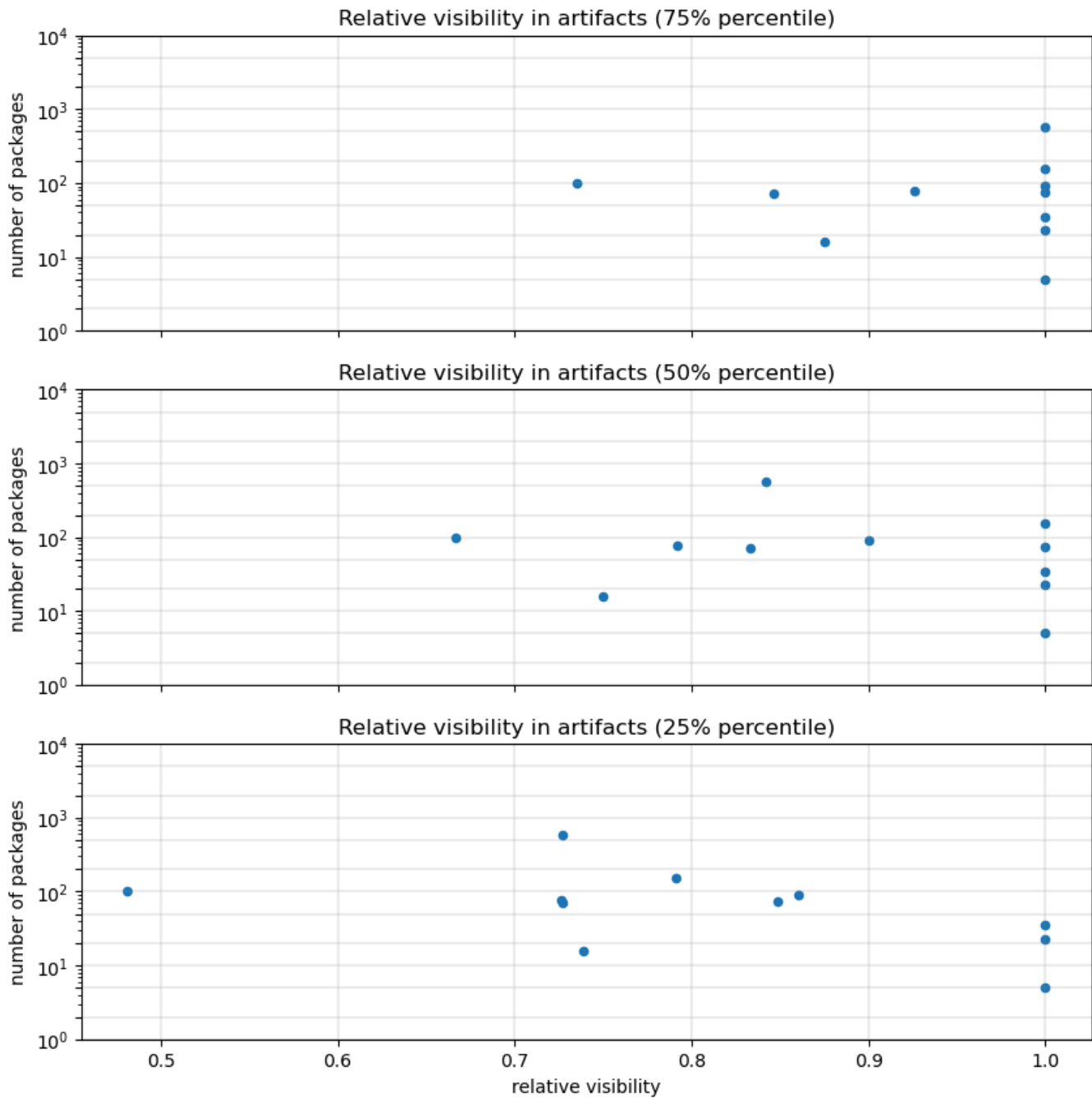


Table 2a - Top 40 packages with the highest visibility and lowest encapsulation

This table shows the relative visibility statistics per packages and artifact and focusses on packages with many types, hardly any package protected ones and therefore the highest relative visibility (lowest encapsulation). Package protected types would help to improve encapsulation.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

`Relative_visibility_public_types_to_all_types_per_package`

	artifactName	fullQualifiedPackageName	packageName	publicTypes	allTypes	relativeVisibility
0	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	annotation	18	18	1.0
1	axon-modelling-5.0.3	org.axonframework.modelling	modelling	16	16	1.0
2	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conf...	configuration	14	14	1.0
3	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	child	14	14	1.0
4	axon-common-5.0.3	org.axonframework.common.jdbc	jdbc	12	12	1.0
5	axon-messaging-5.0.3	org.axonframework.messaging.core.sequencing	sequencing	11	11	1.0
6	axon-conversion-5.0.3	org.axonframework.conversion.avro	avro	10	10	1.0
7	axon-common-5.0.3	org.axonframework.common.property	property	9	9	1.0
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	jdbc	9	9	1.0
9	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.tracing	tracing	7	7	1.0
10	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.tracing	tracing	7	7	1.0
11	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.util	util	7	7	1.0
12	axon-update-5.0.3	org.axonframework.update.configuration	configuration	7	7	1.0
13	axon-common-5.0.3	org.axonframework.common.util	util	6	6	1.0
14	axon-messaging-5.0.3	org.axonframework.messaging.tracing.attributes	attributes	6	6	1.0
15	axon-messaging-5.0.3	org.axonframework.messaging.core.correlation	correlation	6	6	1.0
16	axon-update-5.0.3	org.axonframework.update.api	api	6	6	1.0
17	axon-common-5.0.3	org.axonframework.common.lifecycle	lifecycle	5	5	1.0
18	axon-conversion-5.0.3	org.axonframework.conversion.converter	converter	5	5	1.0
19	axon-conversion-5.0.3	org.axonframework.conversion.jackson	jackson	5	5	1.0
20	axon-conversion-5.0.3	org.axonframework.conversion.jackson2	jackson2	5	5	1.0
21	axon-messaging-5.0.3	org.axonframework.messaging.monitoring.interce...	interception	5	5	1.0
22	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	store	5	5	1.0
23	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	subscribing	5	5	1.0
24	axon-tracing-opentelemetry-5.0.3	org.axonframework.extension.tracing.opentelemetry	opentelemetry	5	5	1.0
25	axon-update-5.0.3	org.axonframework.update	update	5	5	1.0
26	axon-common-5.0.3	org.axonframework.common.jpa	jpa	4	4	1.0
27	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	jpa	4	4	1.0
28	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.replay	replay	4	4	1.0
29	axon-update-5.0.3	org.axonframework.update.detection	detection	4	4	1.0
30	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	errorhandling	3	3	1.0
31	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	processing	3	3	1.0
32	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration...	reflection	3	3	1.0
33	axon-common-5.0.3	org.axonframework.common.function	function	2	2	1.0
34	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	configuration	2	2	1.0
35	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conv...	conversion	2	2	1.0
36	axon-messaging-5.0.3	org.axonframework.messaging.core.conversion	conversion	2	2	1.0
37	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.gateway	gateway	2	2	1.0
38	axon-messaging-5.0.3	org.axonframework.messaging.core.reflection	reflection	2	2	1.0
39	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.actuato...	axonserver	2	2	1.0

Table 2b - Top 40 packages with the lowest visibility and highest encapsulation

This table shows the relative visibility statistics per packages and artifact and focusses on packages with many types, many package protected ones and therefore the lowest relative visibility (highest encapsulation). Package protected types help to improve encapsulation. Zero percent visibility and therefore packages with no public visible type are suspicious to be dead code.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Relative_visibility_public_types_to_all_types_per_package

	artifactName	fullQualifiedPackageName	packageName	publicTypes	allTypes	relativeVisibility
0	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore.inm...	inmemory	1	5	0.200000
1	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	pooled	6	24	0.250000
2	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.inte...	interception	2	8	0.250000
3	axon-messaging-5.0.3	org.axonframework.messaging.monitoring.configu...	configuration	3	9	0.333333
4	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	inmemory	1	3	0.333333
5	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore.jpa	jpa	6	13	0.461538
6	axon-messaging-5.0.3	org.axonframework.messaging.core.interception	interception	10	21	0.476190
7	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.configuration	configuration	7	14	0.500000
8	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.in...	interception	3	6	0.500000
9	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	annotation	1	2	0.500000
10	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.an...	annotation	1	2	0.500000
11	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	jdbc	1	2	0.500000
12	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	jpa	1	2	0.500000
13	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.autoconfig	autoconfig	21	41	0.512195
14	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	9	16	0.562500
15	axon-test-5.0.3	org.axonframework.test	test	3	5	0.600000
16	axon-common-5.0.3	org.axonframework.common.caching	caching	8	13	0.615385
17	axon-messaging-5.0.3	org.axonframework.messaging.core	core	50	80	0.625000
18	axon-common-5.0.3	org.axonframework.common.infra	infra	5	8	0.625000
19	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	event	9	14	0.642857
20	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation.ref...	reflection	4	6	0.666667
21	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.inte...	interception	2	3	0.666667
22	axon-modelling-5.0.3	org.axonframework.modelling.configuration	configuration	9	13	0.692308
23	axon-conversion-5.0.3	org.axonframework.conversion	conversion	7	10	0.700000
24	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.repl...	annotation	7	10	0.700000
25	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing	eventsourcing	5	7	0.714286
26	axon-test-5.0.3	org.axonframework.test.fixture	fixture	23	32	0.718750
27	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.an...	annotation	8	11	0.727273
28	axon-messaging-5.0.3	org.axonframework.messaging.core.interception....	annotation	8	11	0.727273
29	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.dist...	distributed	8	11	0.727273
30	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer	micrometer	8	11	0.727273
31	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.query	query	8	11	0.727273
32	axon-messaging-5.0.3	org.axonframework.messaging.core.timeout	timeout	6	8	0.750000
33	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	token	6	8	0.750000
34	axon-common-5.0.3	org.axonframework.common.annotation	annotation	3	4	0.750000
35	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.co...	configuration	3	4	0.750000
36	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.conf...	configuration	3	4	0.750000
37	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer...	springboot	3	4	0.750000
38	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.service...	connection	3	4	0.750000
39	axon-test-5.0.3	org.axonframework.test.extension	extension	3	4	0.750000

Table 2 Chart 1 - Relative visibility of packages

```
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Parameters 'cmap' will be ignored
scatter = ax.scatter(
```

<Figure size 640x480 with 0 Axes>

